ECE 332 — Homework 4: Generated Answers from HW4 Cheat Sheet

Each problem: state the goal equation → identify missing variables → solve each → plug back → box the answer.

Symbol reference (used throughout):

- l — antenna physical length (m)
- $\lambda = c/f$ — wavelength (m); $c = 3\times10^8$ m/s
- I_0 — peak current at antenna feed (A)
- R — distance from antenna to observation point (m)
- θ — angle from antenna axis (rad or deg)
- $\eta_0 = 120\pi \approx 377\,\Omega$ — intrinsic impedance of free space
- $k = 2\pi/\lambda$ — wavenumber (rad/m)
- $S(R,\theta)$ — average power density radiated (W/m^2)
- $F(\theta) = S(\theta)/S_{\max}$ — normalized radiation intensity (unitless)
- β — half-power beamwidth (rad or deg)
- Ω_p — pattern solid angle (sr)
- $D = 4\pi/\Omega_p$ — directivity (unitless, often dB)
- $A_e = \lambda^2 D/(4\pi)$ — effective aperture area (m^2)
- A_p — physical cross-section area (m^2)
- G_t, G_r — antenna gains (linear)
- P_t, P_r — transmitted / received power (W)
- $k_B = 1.38\times10^{-23}$ J/K — Boltzmann constant
- T_{sys} — system noise temperature (K)
- B — receiver bandwidth (Hz)

Key shortcuts: *Hertzian dipole ($l/\lambda < 1/50$) ⇒ $D = 1.5$. Half-wave dipole ($l = \lambda/2$) ⇒ $D = 1.64$, $R_{rad} = 73.2\,\Omega$. 3 dB ⇒ linear gain of 2.*

Problem 1 — Hertzian Dipole Power Density at 5 km — 8 pts

Setup: $l = 1$ m, $f = 1$ MHz, $I_0 = 12$ A, $R = 5$ km, $\theta = 45°$. Find $S(R,\theta)$.

DIPOLE TYPE BY LENGTH

Compute first *$\lambda = c/f$, then ratio l/λ*

l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R,\theta)$, $D = 1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave short-cut, $D = 1.64$, $R_{\text{rad}} = 73\,\Omega$
$= 1$	Full-wave	ARB, $D = 2.41$, $R_{\text{rad}} = 199\,\Omega$
other	Arbitrary	ARB formula

If $l/\lambda < 1/50$ STOP and use Hertzian. Do NOT plug small l into the arbitrary formula — wrong shape.

COMMON DIPOLE PARAMETERS

Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
$\lambda/4$ monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{rad} = 36.6 I_0^2$ W. $\lambda/4$ monopole (above ground): same patt. as half-wave but P_{rad}, R_{rad} halved.
Lossless antenna ($\xi = 1$) $\Rightarrow$ $G = D$. So in Friis, use G_t, G_r = these D values (e.g. half-wave $G = 1.64$).

HERTZIAN DIPOLE — $S(R,\theta)$

Far-field E,H phasors *small $l \ll \lambda$*

$$\tilde{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi}\frac{e^{-jkR}}{R}\sin\theta, \quad \tilde{H}_\phi = \tilde{E}_\theta/\eta_0$$

Power density *$l/\lambda < 1/50$; far field*

$$S(R,\theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}\sin^2\theta = S_{\max}\sin^2\theta\ (\text{W/m}^2)$$

$S(R,\theta)$ — avg. pwr dens. at (R,θ) (W/m^2)
l — antenna length (m, the rod itself)
I_0 — peak current in dipole (A)
R — distance from antenna to obs. point (m)
θ — angle from dipole axis (rad/°)
$k = 2\pi/\lambda$ — wavenumber (rad/m)
$\eta_0 = 120\pi \approx 377\,\Omega$ — free-space impedance

Max direction *broadside* $\theta_{\max} = \pi/2$

Total radiated power *integrate over 4π*

$$P_{\text{rad}} = \frac{\eta_0\pi}{3}\left(\frac{I_0 l}{\lambda}\right)^2\ (\text{W})$$

Directivity *exact* $D_{\text{Hertz}} = 1.5$ (1.76 dB)

Step 0 — Classify the dipole. From DIPOLE TYPE BY LENGTH, we need l/λ.

$$\lambda = \frac{c}{f} = \frac{3\times10^8}{10^6} = 300\text{ m}, \quad \frac{l}{\lambda} = \frac{1}{300} < \frac{1}{50} \Rightarrow \textbf{Hertzian}$$

Goal: find $S(R,\theta)$ using the Hertzian power density formula (from HERTZIAN DIPOLE box):

$$S(R,\theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2}\sin^2\theta$$

Need: η_0, k, I_0, l, R, θ.

Solve each unknown:

η_0: constant, free-space impedance.

$$\eta_0 = 120\pi\,\Omega$$

k: wavenumber from wavelength.

$$k = \frac{2\pi}{\lambda} = \frac{2\pi}{300}\text{ rad/m}$$

I_0, l, R, θ: given.

$$I_0 = 12\text{ A}, \quad l = 1\text{ m}, \quad R = 5000\text{ m}, \quad \theta = 45°$$

Plug back in:

$$S = \frac{(120\pi)(2\pi/300)^2(12)^2(1)^2}{32\pi^2(5000)^2}\sin^2(45°)$$

With $\sin^2(45°) = 1/2$:

$$S = \frac{(120\pi)(4\pi^2/90000)(144)(1)}{32\pi^2(25\times10^6)}\cdot\frac{1}{2}$$

Simplify: numerator has $120\pi \cdot 4\pi^2 \cdot 144/90000 = 69120\pi^3/90000$, denominator $32\pi^2 \cdot 25\times10^6 = 8\times10^8\pi^2$. Combine and multiply by 1/2:

$$\boxed{S(5 \text{ km}, 45^\circ) = 1.51\times10^{-9} \text{ W/m}^2 = 1.51 \text{ nW/m}^2}$$

Sanity: 12 A at 1 MHz through 1 m radiates 1.5 nW/m^2 at 5 km. The $1/R^2$ falloff dominates — most power scatters into space, only a tiny fraction reaches any one point.

Problem 2 — Beam Parameters from $F(\theta) = e^{-20\theta^2}$ 15 pts

Setup: $F(\theta) = \exp(-20\theta^2)$ for $0 \le \theta \le \pi$ (θ in radians). Find (a) HPBW β, (b) Ω_p, (c) D.

BEAM PARAMETERS — β, Ω_p, D

Step 1 — Normalize *always start here*

$$F(\theta) = S(\theta)/S_{\max} \text{ (unitless)}$$

Step 2 — HPBW β *half-power beamwidth, full angular width at -3 dB*

$$\text{Set } F(\theta) = 0.5, \text{ solve for } \theta \rightarrow \theta_{1/2}$$

$$\beta = 2\theta_{1/2} \text{ (symmetric beams, rad or }^\circ)$$

Gaussian:

$$F(\theta_{1/2}) = e^{-a\theta_{1/2}^2} = 0.5 \Rightarrow \theta_{1/2} = \sqrt{\ln 2/a}$$

Step 3 — Pattern solid angle Ω_p *units: sr*

$$\Omega_p = \int_0^{2\pi}\int_0^{\pi} F(\theta)\sin\theta\, d\theta\, d\phi$$

No ϕ dep.: $\Omega_p = 2\pi\int_0^{\pi} F(\theta)\sin\theta\, d\theta$.

TI-36X Pro: `[2nd][d/dx]` *for* $\int_{\square}^{\square}$*, set RAD mode, multiply result by* 2π.

Narrow-beam Gaussian shortcut ($\sin\theta \approx \theta$, $\rightarrow \infty$)*:*

$$\Omega_p \approx \frac{\pi}{a} = \frac{\pi\,\theta_{1/2}^2}{\ln 2} \quad (F = e^{-a\theta^2})$$

Step 4 — Directivity D

$$D = \frac{4\pi}{\Omega_p} \text{ (unitless)} \qquad D_{\text{dB}} = 10\log_{10} D$$

Alt. with two orthogonal HPBWs: $D \approx 4\pi/(\beta_{xz}\beta_{yz})$.

Step 5 — Gain (if asked) $\xi =$ *rad. eff.*, $0 < \xi \le 1$

$$G = \xi D \qquad G_{\text{dB}} = 10\log_{10} G$$

Lossless / ideal antenna $\Rightarrow G = D$.

(a) Half-power beamwidth β (5 pts)

Goal: find β using the HPBW procedure (BEAM PARAMETERS Step 2):

$$\text{Set } F(\theta) = 0.5, \text{ solve for } \theta \to \theta_{1/2}$$

$$\beta = 2\theta_{1/2} \quad \text{(symmetric beam)}$$

Need: $\theta_{1/2}$, found by setting our specific $F(\theta) = 0.5$.

Solve $\theta_{1/2}$: plug $F = \exp(-20\theta^2)$:

$$\exp(-20\theta_{1/2}^2) = 0.5 \Rightarrow -20\theta_{1/2}^2 = \ln(0.5) = -0.693$$

$$\theta_{1/2}^2 = \frac{0.693}{20} = 0.03466, \quad \theta_{1/2} = \sqrt{0.03466} = 0.186 \text{ rad}$$

That's the half-angle from peak to one -3 dB point. Since the beam is symmetric about $\theta = 0$, the *full* half-power beamwidth is twice that:

$$\boxed{\beta = 2\theta_{1/2} = 0.372 \text{ rad} = 21.31^\circ}$$

(b) Pattern solid angle Ω_p (5 pts)

Goal: find Ω_p using the integral definition (BEAM PARAMETERS Step 3):

$$\Omega_p = \iint_{4\pi} F(\theta)\sin\theta\, d\theta\, d\phi$$

Need: evaluate this integral with our specific $F(\theta) = e^{-20\theta^2}$.

Set up the integral: F has no ϕ dependence, so $\int_0^{2\pi} d\phi = 2\pi$:

$$\Omega_p = 2\pi \int_0^{\pi} e^{-20\theta^2} \sin\theta\, d\theta$$

Solve numerically on TI-36X Pro (no closed form):

1. `[mode]` $\rightarrow$ **RAD**
2. `[2nd][d/dx]` for $\int_{\square}^{\square}$ template
3. Lower limit: `0`; Upper limit: `[`π`]`
4. Integrand: `e^(-20×x^2)×sin(x)`
5. Variable: `x`; `[enter]` $\rightarrow$ returns ≈ 0.024792
6. Multiply: `×[`π`]×2 [enter]` $\rightarrow \approx 0.15578$

$$\boxed{\Omega_p \approx 0.1558 \text{ sr}}$$

Keep the unrounded value for part (c). Textbook rounds to 0.156 *first and gets* $D = 80.55$*; using full precision gives* $D \approx 80.67$.

(c) Directivity D (5 pts)

Goal: find D using the standard formula (BEAM PARAMETERS Step 4):

$$D = \frac{4\pi}{\Omega_p}$$

Need: Ω_p (from part b).

Plug in: $\Omega_p = 0.15578$ sr:

$$D = \frac{4\pi}{0.15578} = \frac{12.566}{0.15578}$$

$$\boxed{D = 80.67}$$

In dB: $D_{dB} = 10\log_{10}(80.67) = 19.07$ *dB.*

Sanity: $D \approx 80$ is highly directive (much narrower than half-wave's $D = 1.64$). The narrow Gaussian-like beam ($\beta \approx 21^\circ$) concentrates radiation into a small solid angle, so directivity is large.

Problem 3 — $\lambda/4$ Dipole Radiation Pattern 15 pts

Setup: Dipole of length $l = \lambda/4$. Find (a) $\theta_{\max}$, (b) $S_{\max}$, (c) plot $F(\theta)$.

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Power density at R *any l*

$$S(\theta) = \frac{15I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right)-\cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2$$

At $\theta=0,\pi$: bracket is 0/0. L'Hôpital $\Rightarrow S=0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta=0.001$ rad $\to$ tiny $\to 0$.

Normalized pattern *dimensionless* $F(\theta)=S(\theta)/S_{\max}$

Quarter-wave $l=\lambda/4$ $\pi l/\lambda=\pi/4$, $\cos(\pi/4)=\sqrt{2}/2$

$$\frac{S(\theta)}{S_0} = \left[\frac{\cos(\frac{\pi}{4}\cos\theta)-\frac{\sqrt{2}}{2}}{\sin\theta}\right]^2, \quad S_0 = \frac{15I_0^2}{\pi R^2}$$

$$S_{\max} = S(\pi/2) = S_0\left(1-\frac{\sqrt{2}}{2}\right)^2 \approx 0.0858\,S_0$$

Half-wave $l=\lambda/2$ $\pi l/\lambda=\pi/2$, *shortcut form*

$$S(\theta) = \frac{15I_0^2}{\pi R^2}\frac{\cos^2(\frac{\pi}{2}\cos\theta)}{\sin^2\theta}$$

$D=1.64$, $R_{\text{rad}}=73\,\Omega$, $\theta_{\max}=\pi/2$.

COMMON DIPOLE PARAMETERS

Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
$\lambda/4$ monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{rad}=36.6I_0^2$ W. $\lambda/4$ monopole (above ground): same patt. as half-wave but P_{rad}, R_{rad} halved.
Lossless antenna ($\xi = 1$) $\Rightarrow G = D$. So in Friis, use G_t, G_r = these D values (e.g. half-wave $G=1.64$).

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

°	rad	$\sin\theta$	$\cos\theta$	$\sin^2\theta$	$\cos^2\theta$
0	0	0	1	0	1
30	$\pi/6$	$1/2$	$\sqrt{3}/2$	$1/4$	$3/4$
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	$1/2$	$1/2$
60	$\pi/3$	$\sqrt{3}/2$	$1/2$	$3/4$	$1/4$
90	$\pi/2$	1	0	1	0
180	π	0	-1	0	1

Convert: $\theta_{rad} = \theta_{deg}\cdot\pi/180$. Antenna formulas usually take θ in rad.

Common setup for all 3 parts.

Goal: starting point for parts (a)-(c) is the arbitrary-length dipole power density (from ARBITRARY-LENGTH DIPOLE box):

$$S(\theta) = \frac{15I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right)-\cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2$$

Need: plug in $l = \lambda/4$ to simplify.

Substitute $l = \lambda/4 \Rightarrow \pi l/\lambda = \pi/4$:

$$S(\theta) = \frac{15I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi}{4}\cos\theta\right)-\cos(\pi/4)}{\sin\theta}\right]^2$$

From COMMON ANGLES table: $\cos(\pi/4) = \sqrt{2}/2 = 0.7071$.

$$S(\theta) = S_0\left[\frac{\cos\left(\frac{\pi}{4}\cos\theta\right)-\frac{\sqrt{2}}{2}}{\sin\theta}\right]^2, \quad S_0 \equiv \frac{15I_0^2}{\pi R^2}$$

(a) Direction of maximum radiation $\theta_{\max}$ (5 pts)

Goal: find $\theta_{\max}$ where $S(\theta)$ is maximum. **Need:** know the shape of $S(\theta)$ over $\theta \in [0, \pi]$. Three-step argument: endpoints, symmetry, sampling.

Step (i) — endpoints via L'Hôpital. Define $f(\theta) = \cos\left(\frac{\pi}{4}\cos\theta\right) - \frac{\sqrt{2}}{2}$ and $g(\theta) = \sin\theta$. At $\theta = 0$:

$$f(0) = \cos(\pi/4 \cdot 1) - \sqrt{2}/2 = \sqrt{2}/2 - \sqrt{2}/2 = 0$$

$$g(0) = \sin(0) = 0 \Rightarrow 0/0 \text{ form, apply L'Hôpital.}$$

Derivatives (chain rule for f):

$$f'(\theta) = \tfrac{\pi}{4}\sin\left(\tfrac{\pi}{4}\cos\theta\right)\sin\theta, \quad g'(\theta) = \cos\theta$$

$$\lim_{\theta \to 0} \tfrac{f'(\theta)}{g'(\theta)} = \tfrac{(\pi/4)\sin(\pi/4)\cdot 0}{1} = 0$$

So $S(0)/S_0 = 0^2 = 0$. Same algebra at $\theta = \pi$: $S(\pi) = 0$. **Peak is in** $(0, \pi)$**.**

Step (ii) — symmetry. Substitute $\theta \to \pi - \theta$: $\cos(\pi - \theta) = -\cos\theta$, but cos is even so $\cos((\pi/4)\cos(\pi - \theta)) = \cos((\pi/4)\cos\theta)$. And $\sin(\pi - \theta) = \sin\theta$. So $S(\theta) = S(\pi - \theta)$. **Pattern mirrors about** $\theta = \pi/2$**.**

Step (iii) — sample to confirm single peak.

At $\theta = 30^\circ$: $\cos(30^\circ) = 0.866$, $(\pi/4)(0.866) = 0.680$, $\cos(0.680) = 0.777$, $0.777 - 0.707 = 0.070$, $\sin(30^\circ) = 0.500$, $0.070/0.500 = 0.140$, $(0.140)^2 = \mathbf{0.020}$.

At $\theta = 60^\circ$: $\cos(60^\circ) = 0.500$, $(\pi/4)(0.500) = 0.393$, $\cos(0.393) = 0.924$, $0.924 - 0.707 = 0.217$, $\sin(60^\circ) = 0.866$, $0.217/0.866 = 0.250$, $(0.250)^2 = \mathbf{0.063}$.

At $\theta = 90^\circ$: $\cos(90^\circ) = 0$, $(\pi/4)(0) = 0$, $\cos(0) = 1$, $1 - 0.707 = 0.293$, $\sin(90^\circ) = 1$, $0.293/1 = 0.293$, $(0.293)^2 = \mathbf{0.0858}$.

θ	30°	60°	90°
S/S_0	0.020	0.063	**0.0858**

Monotone rise to 90°, then symmetric fall (by step (ii)). Single peak at the center.

$$\boxed{\theta_{\max} = \pi/2 = 90^\circ \text{ (broadside)}}$$

(b) Expression for $S_{\max}$ (5 pts)

Goal: find $S_{\max}$ by evaluating S at the maximum direction:

$$S_{\max} = S(\theta_{\max})$$

Need: $\theta_{\max}$ (from part a: $\theta_{\max} = \pi/2$).

Plug $\theta = \pi/2$ into $S(\theta)$ step by step:

Inner cos: $\cos(\pi/2) = 0$, so $\frac{\pi}{4}\cos(\pi/2) = 0$.

Outer cos: $\cos(0) = 1$.

Denominator: $\sin(\pi/2) = 1$.

Bracket: $(1 - \sqrt{2}/2)/1 = 1 - \sqrt{2}/2 = 1 - 0.7071 = 0.2929$.

Square: $(0.2929)^2 = 0.0858$.

Multiply by S_0:

$$\boxed{S_{\max} = 0.0858 \cdot \frac{15 I_0^2}{\pi R^2} \approx \frac{1.287\, I_0^2}{\pi R^2}}$$

(c) Normalized pattern $F(\theta) = S(\theta)/S_{\max}$ (5 pts)

Goal: find $F(\theta)$ by dividing $S(\theta)$ by $S_{\max}$, then plot. **Need:** $S(\theta)$ and $S_{\max}$ (both from above).

Compute the normalization coefficient: $1/0.0858 = 11.66$.

Form $F(\theta)$:

$$\boxed{F(\theta) = 11.66 \left[\frac{\cos\!\left(\frac{\pi}{4}\cos\theta\right) - \frac{\sqrt{2}}{2}}{\sin\theta}\right]^2}$$

Plot: bell-shaped curve peaking at $\theta = \pi/2$ with $F = 1$, dropping to 0 at $\theta = 0$ and π. Symmetric about $\theta = \pi/2$. Similar shape to half-wave dipole pattern but slightly narrower since the dipole is shorter.

Problem 4 — Effective Area vs. Physical Cross-section 5 pts

Setup: Half-wave dipole at $f = 100$ MHz, wire diameter $d = 2$ cm. Find A_e and compare to A_p.

EFFECTIVE AREA A_e

Definition *antenna's RX cross section (m^2)*

$$A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}$$

Physical cross section *wire dipole, diameter d*

$$A_p = l\,d = \frac{\lambda}{2}\,d \text{ (half-wave)}$$

Inverse: D from A_e

$$D = \frac{4\pi A_e}{\lambda^2} \text{ (unitless)}$$

Hertzian special case $D = 1.5$

$$A_e^{\text{Hertz}} = \frac{3\lambda^2}{8\pi} \approx 0.119\lambda^2 \text{ (m}^2\text{)}$$

Thin wire: $A_e \gg A_p$. Antenna captures field over $\sim\lambda$.

COMMON DIPOLE PARAMETERS

Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2(l/\lambda)^2$
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Full-wave	λ	2.41	3.82	199

Half-wave: $P_{rad} = 36.6 I_0^2$ W. $\lambda/4$ monopole (above ground): same patt. as half-wave but P_{rad}, R_{rad} halved.
Lossless antenna ($\xi = 1$) $\Rightarrow$ $G = D$. So in Friis, use G_t, G_r = these D values (e.g. half-wave $G = 1.64$).

Part 1 — Effective area A_e

Goal: find A_e using the standard formula (from EFFECTIVE AREA box):

$$A_e = \frac{\lambda^2 D}{4\pi}$$

Need: λ and D.

Solve each unknown:

λ: use $\lambda = c/f$.

$$\lambda = \frac{c}{f} = \frac{3\times10^8}{10^8} = 3 \text{ m}$$

D: look up half-wave in COMMON DIPOLE PARAMETERS table.

$$D_{\text{half-wave}} = 1.64$$

Plug back in:

$$A_e = \frac{(3)^2(1.64)}{4\pi} = \frac{9\cdot1.64}{4\pi} = \frac{14.76}{12.566}$$
$$\boxed{A_e = 1.17 \text{ m}^2}$$

Part 2 — Physical cross-section A_p

Goal: find A_p using the wire-dipole formula:

$$A_p = l \cdot d$$

Need: l and d.

Solve each unknown:

l: half-wave means $l = \lambda/2$ (from COMMON DIPOLE PARAMS table).

$$l = \frac{\lambda}{2} = \frac{3}{2} = 1.5 \text{ m}$$

d: given.

$$d = 2 \text{ cm} = 0.02 \text{ m}$$

Plug back in:

$$A_p = (1.5)(0.02) = \boxed{0.03 \text{ m}^2}$$

Part 3 — Ratio

$$\frac{A_e}{A_p} = \frac{1.17}{0.03} = \boxed{39.2}$$

Sanity: the antenna's "catching area" is 39× its physical wire cross-section. A thin wire interacts with the field over a region $\sim \lambda$ around itself, so it captures much more energy than its tiny wire would suggest.

Problem 5 — Friis Transmission, TV Broadcast 5 pts

Setup: Half-wave dipole TX, $P_t = 1$ kW at $f = 50$ MHz. RX antenna with $G_r = 3$ dB, distance $R = 30$ km. Find P_r.

FRIIS TRANSMISSION FORMULA

Far-field link budget *P_t tx, P_r rx, range R*

$$\boxed{P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t} \text{ (W)}$$

G_t, G_r are linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.

Solve for R *max range*

$$R = \frac{\lambda}{4\pi}\sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e *recv. area form*

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) *when patterns NOT peak-aligned*

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t \, F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert dB gains to linear: $G = 10^{G_{\mathrm{dB}}/10}$.
3. If TX is "half-wave dipole" use $G_t = 1.64$. 4. Plug into Friis.
Use linear G, not dB. Convert FIRST.

COMMON DIPOLE PARAMETERS

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dB ↔ LINEAR

$$G_{\mathrm{dB}} = 10\log_{10} G_{\mathrm{lin}} \qquad G_{\mathrm{lin}} = 10^{G_{\mathrm{dB}}/10}$$

dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
-3	0.5	-10	0.1

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.
For E-field / amplitude use 20 log not 10 log (power $\propto E^2$).

Goal: find P_r using the Friis transmission formula (from FRIIS TRANSMISSION FORMULA box):

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t$$

Need: G_t, G_r (both linear), λ, R, P_t.

Solve each unknown:

λ: $\lambda = c/f$.

$$\lambda = \frac{3\times10^8}{5\times10^7} = 6 \text{ m}$$

G_t: TX is half-wave dipole, COMMON DIPOLE PARAMS gives $D = 1.64$, and for lossless antenna $G = D$.

$$G_t = 1.64$$

G_r: given as 3 dB. Convert to linear via dB → LINEAR table: 3 dB = ×2.

$$G_r = 10^{3/10} = 2$$

R, P_t: given.

$$R = 30 \text{ km} = 3\times10^4 \text{ m}, \quad P_t = 10^3 \text{ W}$$

Plug back in: compute the path-loss factor first.

$$\tfrac{\lambda}{4\pi R} = \tfrac{6}{4\pi(3\times10^4)} = \tfrac{6}{3.77\times10^5} = 1.59\times10^{-5}$$

$$\left(\tfrac{\lambda}{4\pi R}\right)^2 = (1.59\times10^{-5})^2 = 2.53\times10^{-10}$$

$$P_r = (1.64)(2)(2.53\times10^{-10})(10^3) = (3.28)(2.53\times10^{-7})$$

$$\boxed{P_r = 8.3\times10^{-7} \text{ W} = 0.83\ \mu\text{W}}$$

Sanity: 1 kW in, 0.83 μW out at 30 km. A 90 dB drop, dominated by $1/R^2$ path loss. Plenty for a TV receiver (sensitivity $\sim 10^{-13}$ W).

Problem 6 — Communication Link with SNR 15 pts (new)

Setup: $G_t = 20$ dB, $G_r = 23$ dB, $R = 20$ km, $P_t = 10$ W, $f = 6$ GHz. (c) $T_{\text{sys}} = 1000$ K, $B = 20$ MHz.

FRIIS TRANSMISSION FORMULA

Far-field link budget *P_t tx, P_r rx, range R*

$$\boxed{P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t} \text{ (W)}$$

G_t, G_r are linear gains; $\lambda = c/f$; R in m; P_t, P_r in W.

Solve for R *max range*

$$R = \frac{\lambda}{4\pi}\sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e *recv. area form*

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, so $P_r = G_t G_r P_t / L_{fs}$.

General form (off-axis) *when patterns NOT peak-aligned*

$$P_r = G_t G_r \left(\frac{\lambda}{4\pi R}\right)^2 P_t \, F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

F_t, F_r normalized patterns at the link directions; $F = 1$ when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert dB gains to linear: $G = 10^{G_{\text{dB}}/10}$.
3. If TX is "half-wave dipole" use $G_t = 1.64$. 4. Plug into Friis.

Use linear G, not dB. Convert FIRST.

dB ↔ LINEAR

$$G_{\text{dB}} = 10\log_{10} G_{\text{lin}} \qquad G_{\text{lin}} = 10^{G_{\text{dB}}/10}$$

dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
−3	0.5	−10	0.1

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use $20\log$ not $10\log$ (power $\propto E^2$).

NOISE & SNR

Noise power *thermal noise into matched receiver*

$$P_N = k_B \, T_{\text{sys}} \, B \text{ (W)}$$

$k_B = 1.38\times10^{-23}$ J/K; T_{sys} system noise temp (K); B receiver bandwidth (Hz).

Signal-to-noise ratio

$$\text{SNR} = \frac{P_{\text{rec}}}{P_N} \text{ (unitless)} \qquad \text{SNR}_{\text{dB}} = 10\log_{10}\frac{P_{\text{rec}}}{P_N}$$

Typical comm. link needs SNR > 10 dB for usable signal.

Shared setup: convert dB to linear and compute λ.

$$\lambda = \frac{c}{f} = \frac{3\times10^8}{6\times10^9} = 0.05 \text{ m}$$

$$G_t = 10^{20/10} = 100, \quad G_r = 10^{23/10} = 199.5$$

(a) Power density at RX antenna S_r (5 pts)

Goal: find S_r using power-spreading-over-sphere (with TX gain factored in):

$$S_r = \frac{G_t P_t}{4\pi R^2}$$

Need: G_t, P_t, R.

Plug in:

$$S_r = \frac{(100)(10)}{4\pi(2\times10^4)^2} = \frac{1000}{4\pi(4\times10^8)} = \frac{1000}{5.027\times10^9}$$

$$\boxed{S_r = 1.989\times10^{-7} \text{ W/m}^2 \approx 0.199 \; \mu\text{W/m}^2}$$

(b) Received power P_r (5 pts)

Goal: find P_r using Friis (from FRIIS TRANSMISSION box):

$$P_r = G_t G_r \left(\tfrac{\lambda}{4\pi R}\right)^2 P_t$$

Need: G_t, G_r, λ, R, P_t (all already converted).

Plug in:

$$\tfrac{\lambda}{4\pi R} = \tfrac{0.05}{4\pi(2\times 10^4)} = 1.989\times 10^{-7}$$

$$\left(\tfrac{\lambda}{4\pi R}\right)^2 = 3.957\times 10^{-14}$$

$$P_r = (100)(199.5)(3.957\times 10^{-14})(10) = 7.894\times 10^{-9}$$

$$\boxed{P_r = 7.89\text{ nW}}$$

Cross-check via effective area: $A_{e,r} = G_r\lambda^2/(4\pi) = 199.5(0.0025)/(4\pi) = 0.0397\text{ m}^2$, then $P_r = S_r A_{e,r} = (1.989\times 10^{-7})(0.0397) = 7.89\times 10^{-9}$ W ✓.

(c) Signal-to-noise ratio in dB (5 pts)

Goal: find SNR in dB. From NOISE & SNR box, two steps:

$$P_N = k_B\, T_{\text{sys}}\, B \quad \text{then} \quad \text{SNR}_{\text{dB}} = 10\log_{10}\tfrac{P_r}{P_N}$$

Need: P_N first ($\rightarrow$ need k_B, T_{sys}, B), then P_r (from part b).

Solve each unknown:

k_B: Boltzmann constant.

$$k_B = 1.38\times 10^{-23}\text{ J/K}$$

T_{sys}, B: given.

$$T_{\text{sys}} = 1000\text{ K}, \quad B = 20\text{ MHz} = 2\times 10^7\text{ Hz}$$

Compute P_N:

$$P_N = (1.38\times 10^{-23})(1000)(2\times 10^7) = 2.76\times 10^{-13}\text{ W}$$

Compute SNR:

$$\text{SNR}_{\text{lin}} = \tfrac{P_r}{P_N} = \tfrac{7.89\times 10^{-9}}{2.76\times 10^{-13}} = 2.86\times 10^4$$

$$\boxed{\text{SNR}_{\text{dB}} = 10\log_{10}(2.86\times 10^4) = 44.6\text{ dB}}$$

Sanity: >40 dB SNR is a very clean link. Plenty for digital comm.

Problem 7 — Circular Aperture Antenna Scaling 15 pts (new)

Setup: Circular aperture, circular beam $\beta = 3°$ at $f = 20$ GHz. (a) D in dB. (b) New D and β if A_p doubled. (c) New D and β if frequency doubled (same A_p).

APERTURE ANTENNAS (horn, dish)

Directivity from aperture area *uniform illumination*

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \quad \Rightarrow \quad A_e \approx A_p$$

Directivity from beamwidths *rectangular beam*

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \text{ (use rad)}$$

Circular beam ($\beta_{xz} = \beta_{yz} = \beta$): $D \approx 4\pi/\beta^2$.

Half-power beamwidth *uniform rect. illumination*

$$\beta_{xz} \approx 0.88\,\frac{\lambda}{\ell_x} \text{ (rad)} \quad \beta_{yz} \approx 0.88\,\frac{\lambda}{\ell_y}$$

Scaling: $D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$. *Double area* $\Rightarrow D\times 2$, $\beta/\sqrt{2}$. *Double* f $(\lambda/2) \Rightarrow D\times 4$, $\beta/2$.

dB ↔ LINEAR

$$G_{\text{dB}} = 10\log_{10} G_{\text{lin}} \qquad G_{\text{lin}} = 10^{G_{\text{dB}}/10}$$

dB	linear	dB	linear
0	1	10	10
3	2	13	20
6	4	20	100
7	5	30	1000
−3	0.5	−10	0.1

Memorize: 3 dB $\Leftrightarrow \times 2$, *10 dB* $\Leftrightarrow \times 10$. *Add dB = multiply linear.*

For E-field / amplitude use $20\log$ *not* $10\log$ *(power* $\propto E^2$*).*

(a) Directivity in dB (5 pts)

Goal: find D using the circular-beam formula (from APERTURE ANTENNAS box):

$$D = \frac{4\pi}{\beta^2} \quad (\beta \text{ in radians})$$

Need: β in radians (given as degrees).

Convert β to radians:

$$\beta = 3° \cdot \frac{\pi}{180} = \frac{\pi}{60} = 0.05236 \text{ rad}$$

Plug in:

$$D = \frac{4\pi}{(0.05236)^2} = \frac{12.566}{0.002742} = 4584$$

$$\boxed{D = 10\log_{10}(4584) = 36.6 \text{ dB}}$$

(b) Area doubled $A_p \to 2A_p$ (5 pts)

Goal: find new D and β using the scaling rules (from APERTURE ANTENNAS box coral note):

$$D \propto A_p \;\Rightarrow\; D_{\text{new}} = 2D_{\text{old}}$$

$$\beta \propto 1/\sqrt{A_p} \;\Rightarrow\; \beta_{\text{new}} = \beta_{\text{old}}/\sqrt{2}$$

Need: D_{old} and β_{old} (from part a).

Plug in:

$$D_{\text{new}} = 2(4584) = 9168 \;\Rightarrow\; \boxed{D_{\text{new}} = 39.6 \text{ dB}}$$

$$\beta_{\text{new}} = 3°/\sqrt{2} = 3°/1.414 = \boxed{2.12°}$$

(c) Frequency doubled $f \to 2f$ $(\lambda \to \lambda/2)$, same A_p (5 pts)

Goal: find new D and β using scaling: from $D = 4\pi A_p/\lambda^2$ we have $D \propto 1/\lambda^2$. Halving λ quadruples D. And $\beta \propto \lambda$, so halving λ halves β. **Need:** D_{old} and β_{old} (from part a).

Plug in:

$$D_{\text{new}} = 4D_{\text{old}} = 4(4584) = 18336 \;\Rightarrow\; \boxed{D_{\text{new}} = 42.6\text{ dB}}$$

$$\beta_{\text{new}} = \beta_{\text{old}}/2 = 3^\circ/2 = \boxed{1.5^\circ}$$

Sanity: bigger area or shorter wavelength both sharpen the beam. Doubling area: +3 dB (×2). Doubling frequency: +6 dB (×4) because both $D \propto A_p/\lambda^2$ and $1/\beta^2 \propto D$.

Problem 8 — 5-Element Linear Array, $d = 3\lambda/4$ (new)

Setup: Broadside linear array, $N = 5$ elements, equal-phase, uniform amplitude, spacing $d = 3\lambda/4$. Find (a) normalized array factor, (b) HPBW.

LINEAR ANTENNA ARRAY

Setup *N identical elements, spacing d along axis*

element pos $z_i = id$, $i = 0, \ldots, N-1$; $A_i = a_i e^{j\psi_i}$

Pattern multiplication

$$S(R, \theta, \phi) = S_e(R, \theta, \phi)\, F_a(\theta)$$

Array factor $F_a(\theta) = |\sum A_i e^{jkid\cos\theta}|^2$

$$F_a(\theta) = \left| \sum_{i=0}^{N-1} A_i\, e^{jkid\cos\theta} \right|^2$$

Uniform $A_i = 1$, equal phase *closed form*

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}, \quad \gamma = kd\cos\theta$$

Normalize: F_a/N^2. Peak N^2 at $\gamma = 0$ ($\cos\theta = 0$, broadside).

HPBW (broadside, uniform) *array length $L = (N-1)d$*

$$\beta \approx 0.88\,\frac{\lambda}{L} \text{ (rad)}$$

(a) Normalized array factor $F_a^{\text{norm}}(\theta)$ (5 pts)

Goal: find $F_a^{\text{norm}}(\theta)$ using the uniform-array closed form (from LINEAR ANTENNA ARRAY box):

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}, \quad \gamma = kd\cos\theta$$

Need: N, k, d, then form γ, then normalize by peak N^2.

Solve each unknown:

N: given.

$$N = 5$$

k: wavenumber.

$$k = \frac{2\pi}{\lambda}$$

d: given.

$$d = \frac{3\lambda}{4}$$

γ: combine.

$$\gamma = kd\cos\theta = \frac{2\pi}{\lambda} \cdot \frac{3\lambda}{4}\cos\theta = \frac{3\pi}{2}\cos\theta$$

So $N\gamma/2 = (5/2)(3\pi/2)\cos\theta = (15\pi/4)\cos\theta$ and $\gamma/2 = (3\pi/4)\cos\theta$.

Plug into F_a:

$$F_a(\theta) = \frac{\sin^2((15\pi/4)\cos\theta)}{\sin^2((3\pi/4)\cos\theta)}$$

Peak occurs at $\gamma = 0$ ($\cos\theta = 0$, broadside $\theta = \pi/2$): $F_a^{\max} = N^2 = 25$.

Normalize:

$$\boxed{F_a^{\text{norm}}(\theta) = \frac{1}{25}\,\frac{\sin^2((15\pi/4)\cos\theta)}{\sin^2((3\pi/4)\cos\theta)}}$$

(b) Half-power beamwidth β (3 pts)

Goal: find β using the uniform-broadside-array HPBW formula:

$$\beta \approx 0.886\frac{\lambda}{Nd} \quad \text{(rad)}$$

Need: N, d, λ.

Plug in:

$$Nd = 5 \cdot \frac{3\lambda}{4} = \frac{15\lambda}{4}$$

$$\beta = 0.886 \cdot \frac{\lambda}{15\lambda/4} = 0.886 \cdot \frac{4}{15} = 0.236 \text{ rad}$$

$$\boxed{\beta = 0.236 \text{ rad} = 13.5^\circ}$$

Cross-check by solving $F_a^{norm} = 0.5$ numerically: get $\gamma/2 \approx 0.28$ rad, so $\cos\theta = 0.28/(3\pi/4) = 0.119$, $\theta = 83.2^\circ$, offset from broadside (90°) by 6.8°, full HPBW $\approx 13.6^\circ$ ✓.

Sanity: 5 elements give a moderately narrow beam ($\sim 14^\circ$) vs. a single dipole's $\sim 78^\circ$. Adding more elements or wider spacing narrows further (with grating-lobe risk if $d > \lambda$).